

An Analysis of Curse of dimensionality

Curse of dimensionality

$$V_d(r) = \frac{\pi^{d/2}}{\Gamma(d/2 + 1)} r^d$$

Curse of dimensionality -- Part 1

Blessing of dimensionality Despite the expected "curse of dimensionality" difficulties, common-sense heuristics based on the most straightforward methods "can yield results which are almost surely optimal" for high-dimensional problems. The term "blessing of dimensionality" was introduced in the late 1990s. Donoho in his "Millennium manifesto" explained why he thinks the "blessing of dimensionality" will form a basis of future data mining. The effects of the blessing of dimensionality were discovered in many applications and found their foundation in the concentration of measure phenomena. One example of the blessing of dimensionality phenomenon is linear separability of a random point from a large finite random set with high probability even if this set is exponentially large: the number of elements in this random set can grow exponentially with dimension. Moreover, this linear functional can be selected in the form of the simplest linear Fisher discriminant. This separability theorem was proven for a wide class of probability distributions: general uniformly log-concave distributions, product distributions in a cube and many other families (reviewed recently in). "The blessing of dimensionality and the curse of dimensionality are two sides of the same coin." For example, the typical property of essentially high-dimensional probability distributions in a high-dimensional space is: the squared distance of random points to a selected point is, with high probability, close to the average (or median) squared distance. This property significantly simplifies the expected geometry of data and indexing of high-dimensional data (blessing), but, at the same time, it makes the similarity search in high dimensions difficult and even useless (curse). Zimek et al. noted that while the typical formalizations of the curse of dimensionality affect i.i.d. data, having data that is separated in each attribute becomes easier even in high dimensions, and argued that the signal-to-noise ratio matters: data becomes easier with each attribute that adds signal, and harder with attributes that only add noise (irrelevant error) to the data. In particular for unsupervised data analysis this effect is known as swamping.

Curse of dimensionality -- Part 2

A common practice of data mining in this domain would be to create association rules between genetic mutations that lead to the development of cancers. To do this, one would have to loop through each genetic mutation of each individual and find other genetic mutations that occur over a desired threshold and create

pairs. They would start with pairs of two, then three, then four until they result in an empty set of pairs. The complexity of this algorithm can lead to calculating all permutations of gene pairs for each individual or row. Given the formula for calculating the permutations of n items with a group size of r is: $n! / (n-r)!$ $\left(\displaystyle \frac{n!}{(n-r)!}\right)$, calculating the number of three pair permutations of any given individual would be 7988004000 different pairs of genes to evaluate for each individual. The number of pairs created will grow by an order of factorial as the size of the pairs increase. The growth is depicted in the permutation table (see right). As we can see from the permutation table above, one of the major problems data miners face regarding the curse of dimensionality is that the space of possible parameter values grows exponentially or factorially as the number of features in the data set grows. This problem critically affects both computational time and space when searching for associations or optimal features to consider. Another problem data miners may face when dealing with too many features is that the number of false predictions or classifications tends to increase as the number of features grows in the data set. In terms of the classification problem discussed above, keeping every data point could lead to a higher number of false positives and false negatives in the model. This may seem counterintuitive, but consider the genetic mutation table from above, depicting all genetic mutations for each individual. Each genetic mutation, whether they correlate with cancer or not, will have some input or weight in the model that guides the decision-making process of the algorithm. There may be mutations that are outliers or ones that dominate the overall distribution of genetic mutations when in fact they do not correlate with cancer. These features may be working against one's model, making it more difficult to obtain optimal results. This problem is up to the data miner to solve, and there is no universal solution. The first step any data miner should take is to explore the data, in an attempt to gain an understanding of how it can be used to solve the problem. One must first understand what the data means, and what they are trying to discover before they can decide if anything must be removed from the data set. Then they can create or use a feature selection or dimensionality reduction algorithm to remove samples or features from the data set if they deem it necessary. One example of such methods is the interquartile range method, used to remove outliers in a data set by calculating the standard deviation of a feature or occurrence.

Curse of dimensionality -- Part 3

The curse of dimensionality refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings such as the three-dimensional physical space of everyday experience. The

An Analysis of Curse of dimensionality

Curse of dimensionality

expression was coined by Richard E. Bellman when considering problems in dynamic programming. The curse generally refers to issues that arise when the number of datapoints is small (in a suitably defined sense) relative to the intrinsic dimension of the data. Dimensionally cursed phenomena occur in domains such as numerical analysis, sampling, combinatorics, machine learning, data mining and databases. The common theme of these problems is that when the dimensionality increases, the volume of the space increases so fast that the available data become sparse. In order to obtain a reliable result, the amount of data needed often grows exponentially with the dimensionality. Also, organizing and searching data often relies on detecting areas where objects form groups with similar properties; in high dimensional data, however, all objects appear to be sparse and dissimilar in many ways, which prevents common data organization strategies from being efficient.